Beth Marie's Ice Cream

Customer Service Time Simulation Report

Monte Carlo Simulation | n = 5,000 Customers

Prepared: April 2026

1. Background & Objective

Beth Marie's Ice Cream collected service-time data on 42 customers, capturing how long each customer took to decide on an order, how long it took staff to scoop or prepare the item, the time spent at the checkout register, and the item type selected (Cone, Cup, Shake, Split, or Sundae).

With only 42 observations, statistical conclusions are limited — small samples are sensitive to individual outliers and may not represent the full range of customer behavior. The goal of this project was to simulate 5,000 customers using the patterns in the original 42 records, creating a dataset large enough for reliable analysis, averages, and comparisons across item types.

2. Original Data Overview

2.1 Sample Size

The raw file contained 50 numbered rows, but rows 43–50 were empty placeholders. The usable dataset consists of 42 complete customer records.

2.2 Variables Collected

- Customer — Sequential ID (1–42)
- Decision Time (ss) — Time in seconds from approaching counter to placing order
- Scoop Time (ss) — Time in seconds to prepare the ice cream item
- Checkout Time (ss) — Time in seconds to complete payment
- Total Service Time (ss) — Scoop Time + Checkout Time
- Item Selected — Category: Cone, Cup, Shake, Split, or Sundae

2.3 Descriptive Statistics (Original, n = 42)

The table below summarizes the key time variables from the real dataset:

Variable	Mean (ss)	Median (ss)	Std Dev	Min (ss)	Max (ss)
Decision Time	40.03	32.78	32.25	8.68	173.16
Scoop Time	31.24	27.73	18.91	9.60	106.70
Checkout Time	20.60	20.50	3.95	13.55	29.45
Total Service Time	51.85	48.00	20.07	23.15	136.15

Key observations:

- Decision Time is the most variable measure — customers ranged from under 9 seconds to over 173 seconds. This large spread reflects a genuine right skew: most customers decide quickly, but occasional indecisive customers pull the average up significantly.
- Scoop Time also shows a wide range (9.6 – 106.7 ss), likely driven by complex items (shakes, splits, sundaes) that require more preparation than a simple cone.
- Checkout Time is the tightest of all variables (std dev only 3.95 ss), suggesting the payment process is highly consistent regardless of what was ordered.
- Total Service Time combines Scoop + Checkout and naturally inherits the variability of Scoop Time.

3. Why Simulate?

Forty-two data points are sufficient to identify patterns and fit a distribution, but they are too few for robust analysis. Common problems with small samples include:

1. Averages are unstable — a single outlier (e.g., the 173-second decision time) can meaningfully shift the mean.
2. Percentile estimates are unreliable — calculating the 90th or 95th percentile from 42 rows gives a rough estimate, not a stable one.
3. Comparisons between item types are unreliable — with only 2 shakes, 1 split, and 1 sundae in the sample, there is not enough data to compare preparation times by item category.
4. Hypothesis testing lacks power — statistical tests require adequate sample sizes to detect meaningful differences.

Simulating 5,000 customers using the same statistical distributions as the real data addresses all of these limitations while remaining faithful to the patterns observed in the original 42 records.

4. Simulation Methodology

4.1 Distribution Selection: Why Lognormal?

Each continuous time variable (Decision Time, Scoop Time, Checkout Time) was simulated using a lognormal distribution. The lognormal was chosen for three reasons:

5. All time values must be positive — normal distributions can generate negative values, which are physically impossible for service times.
6. The data is right-skewed — most customers are served quickly, but a small number take much longer (e.g., a 173-second decision time or a 107-second scoop for a complex order). The lognormal distribution captures this asymmetry naturally.
7. Log-transforming the original values produces an approximately normal distribution, confirming the lognormal fit.

4.2 Fitting the Distributions

For each time variable, the lognormal parameters μ (mu) and σ (sigma) were estimated directly from the 42 real observations using maximum likelihood estimation — specifically, by computing the mean and standard deviation of the natural logarithm of each variable:

Variable	Distribution	μ (log-mean)	σ (log-std)	Rationale
Decision Time	Lognormal	3.4674	0.6532	Right-skewed, outliers present
Scoop Time	Lognormal	3.3139	0.4866	Right-skewed, always positive
Checkout Time	Lognormal	3.0075	0.1917	Tightly clustered, slight skew
Item Selected	Multinomial	—	—	Based on observed proportions
Total Service Time	Derived	—	—	Scoop Time + Checkout Time

These parameters fully describe the shape of each distribution. A lower σ means values cluster tightly around the mean (checkout time), while a higher σ means a wider spread with more extreme values (decision time).

4.3 Item Selection: Multinomial Distribution

The item type (Cone, Cup, Shake, Split, Sundae) was simulated using a multinomial distribution, assigning each simulated customer a randomly drawn item based on the proportions observed in the real data:

Item Type	Orig. Count	Orig. %	Sim. Count	Sim. %
Cone	24	57.1%	2773	55.5%
Cup	14	33.3%	1714	34.3%
Shake	2	4.8%	246	4.9%
Sundae	1	2.4%	142	2.8%
Split	1	2.4%	125	2.5%
Total	42	100%	5000	100%

This ensures the simulated dataset preserves the original product mix.

4.4 Total Service Time

Total Service Time was not simulated independently. Instead, it was calculated as:

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Total Service Time = Scoop Time + Checkout Time
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This matches how the original data was structured and maintains the logical relationship between the variables.

4.5 Random Seed

A fixed random seed (42) was set before generating all simulated values. This ensures reproducibility — running the simulation again will produce the identical 5,000 rows, allowing consistent comparison and verification.

4.6 Software

The simulation was performed in Python using the following libraries:

- numpy — Random number generation (np.random.lognormal, np.random.choice)
- pandas — Data structuring and export
- openpyxl — Excel file creation and formatting

5. Validation: How Well Does the Simulation Match?

The table below compares the key statistics of the original 42 customers to the simulated 5,000, confirming the simulation faithfully reproduces the original patterns:

Variable	Orig. Mean (ss)	Sim. Mean (ss)	Orig. Std Dev	Sim. Std Dev
Decision Time	40.03	39.74	32.25	29.11
Scoop Time	31.24	30.89	18.91	16.20
Checkout Time	20.60	20.66	3.95	4.00
Total Svc Time	51.85	51.56	20.07	16.66

All simulated means are within 1% of the original means, confirming accurate distribution fitting. Standard deviations are slightly narrower in the simulation for Decision and Scoop Time — this is expected with a large sample, as extreme outliers become proportionally less influential while the core distribution shape is preserved.

6. Excel File Structure

The accompanying Excel workbook (BethMaries_Simulated_5000.xlsx) contains three sheets:

- Original Data (50) — The cleaned 42 real customer records from the study.
- Simulated Data (5000) — 5,000 synthetic customer records generated using the lognormal and multinomial distributions described above.
- Summary Statistics — Side-by-side comparison of descriptive statistics for both datasets, plus item distribution breakdown and simulation parameter reference.

7. Limitations & Assumptions

- The simulation assumes all customers are independent — in reality, service times at an ice cream shop may be affected by queue length, time of day, or staff experience, none of which are captured here.

- The lognormal parameters are estimated from only 42 observations, so they carry some uncertainty. A larger original sample would yield more precise parameters.
- Item type was not linked to Scoop Time in the simulation — in reality, shakes and sundaes likely take longer to prepare than cones. Future simulations could model Scoop Time conditional on item type.
- Decision Time was treated as independent of item type, though in practice a customer ordering a complex sundae might take longer to decide.
- The simulation does not model time of day, staffing levels, or seasonal variation.

8. Conclusion

This simulation successfully expanded the 42-customer original dataset into a statistically representative 5,000-customer dataset, preserving the right-skewed shape of service times and the observed product mix. The simulated data is suitable for:

- Estimating reliable percentile benchmarks (e.g., the 90th percentile service time)
- Comparing preparation times across item categories with statistical confidence
- Modeling customer flow and queue behavior in operations research contexts
- Stress-testing service times under different assumptions (e.g., what if shake orders double?)

All simulated values are statistically plausible, grounded in the real observations from Beth Marie's Ice Cream, and reproducible using the documented parameters and Python code.